

COURSE PLAN

(Undergraduate Programme)



F/QSP 11/10/06

Department of Computer Science and Information Technology

Course Code: CS-251

Course Title: Logic Design and Switching Theory

Teacher: Ms. Hera Noor

Semester: Spring

Academic Session: 2026

Class: FSCS (Sec-B & C)

No. of sections: 02

No of contact hours per week per section: 03

TEACHING TOOLS

☒ Problem-based learning	☐ Mini-project	☐ Case studies
☐ Innovative solution	☒ Group activity	☐ Entrepreneurial activities
☐ Industrial visit	☐ Student presentation	☒ Other: <u>Lectures</u>

WEEK No.	TOPICS	NO. OF PERIODS
1.	Introduction to Digital Design, Number Systems, Binary Arithmetic.	03
2.	Basic Logic Gates, and Boolean Algebra.	03
3.	Introduction to Combinational Logic Circuits & Design.	03
4.	Standard (Canonical) forms SOP and POS, Gate level minimization of Boolean Functions using K Map.	03
5.	Active Low Logic and NAND/NOR Implementation.	03
6.	Half Adders, Full Adders, Parallel Adders, 1's Complement Adder/Subtractor, and Carry Look-Ahead Adder.	03
7.	Decoders, Encoders, Multiplexers, and Demultiplexers	03
8.	Mid Term Exam	03
9.	Priority Encoders and BCD to Seven Segment Display Encoders	03
10.	Magnitude Comparator and Parity Checkers	03
11.	Introduction to Sequential Circuits, Latches and Flip flops.	03
12.	Synchronous Counters, and Asynchronous Counters.	03
13.	Shift Registers.	03
14.	Sequential Circuit Design, State Diagram and State tables	03
15.	Switches and Relays	03
Total per Semester:		45

SESSIONAL CRITERIA

Assessment Type	Marks	Schedule (Week No.)
Quiz	10	5,12
Assignment	10	7,14
Mid Term	20	8
Total Sessional Marks	40	

Prepared by: Ms. Hera Noor

Reviewed &
Approved by: Dr. Muhammad Mubashir Khan

Date 28th January 2026

Date _____